Homework2 Sols:

A1 Calculations in S_n:

A1) • $n-1$ Graph(G)

A2) $\langle S_4\rangle = \langle \{(i_0 j_0)(1\,2\cdots n)\}\rangle = S_n$

Aim: $(i_0, i_0+1) \in \langle S_4\rangle$, then

using $\langle (1,2), (1\cdots n)\rangle = S_n$

$g^k \circ g \circ g^{-k} = (\overline{i_0+k}\ \ \overline{j_0+k})$ $\overline{j_0+k} = j_0+k \equiv \pmod n$ and lies in $\{1\cdots n\}$.

pick $k=(j_0-i_0)$, then let $\sigma_s = (\overline{i_0+(s-1)k},\ \overline{i_0+sk})$

§. direct compute

$\sigma_s \cdots \sigma_2\sigma_1\sigma_1\sigma_1^{-1}\cdots \sigma_s^{-1} = (i_0\ \overline{i_0+sk})$ $\rightsquigarrow \exists s\in\mathbb{N}^*.\ \overline{i_0+sk} = i_0+1$ since $(i_0, j_0-i_0) = 1$ $\Rightarrow$ then $\langle S_4\rangle = S_n$.

A3) • $(i,j,k) \overset{\text{conj}}{\sim} (1,2,3)$.

Since if $\sigma\in S_n$. $\sigma(i,j,k)\sigma^{-1} = (1,2,3)$ if • $\sigma\in A_n$ ✓

• $\sigma\notin A_n \Rightarrow \sigma' = \sigma\circ(4,5)$ ✓

A4) • 3-对换.

$\tau = (i,j,k) = (i,k)(i,j)$. pick (s,t) s.t. $\{s,t\}\cap\{i,j,k\} = \emptyset$

$\tau = (s,t)(i,k)\cdot(s,t)(i,j) = (i,k)(i,j) = (i,j,k)$.

(i,k) 属于 A_5

Then since A_5 ~~generated~~ 由3对换生成 $\Rightarrow$ 得证.

A5) • $\text{ord}(x)=2$ "$\Leftarrow$" WLOG. y has fixed point $(1,3)$ $\Rightarrow$ $y=(2,4,5)$ or $y=(2,5,4)$.

y is 3-cycle.

$(1,2)(3,4)\cdot(2,4,5) = (1,2,3,4,5)$. then $\text{ord}(xy)=5$

"$\Rightarrow$" if. y has fixed point $\{1,2\}$ or $\{3,4\}$. $(1,2)(3,4)(2,5,4) = (1,2,5,3,4)$.

$y=(3,4,5)$ or $(3,5,4)$ $\Rightarrow$. $xy(1)=2$ or $xy(2)=1$.

b). $X = \{(u,v) \mid \text{ord}(u)=2,\ \text{ord}(v)=3,\ \text{ord}(uv)=5\}$ $\Rightarrow$. from 1.

u: 不交的2对换 (对)

v: 3 cycle

uv: 5-cycle

u $\{a,b\},\{c,d\}$ then $v=(a,\beta,e)$ $\alpha\in\{a,b\}$ $\beta\in\{c,d\}$

$\frac{1}{2}C_5^2\cdot C_3^2 \cdot 2\times 2\times 2$ (number of u; 2 3-cycles; pick $\{a,b\}$ in; pick in $\{c,d\}$)

so $|X| = \frac{1}{2}C_5^2\cdot C_3^2\cdot 2^3 = 3\times 10\times 8\times\frac{1}{2} = 120$

c) $\text{Aut}(G)\curvearrowright X$ $\text{Aut}(G)\times X \longrightarrow X$.

$(\phi, (u,v)) \longrightarrow (\phi(u)\cdot\phi(v))$

Aim: $\text{Stab}(u,v)=1$ $\forall (u,v)\in X$ means $\langle (u,v)\rangle = G$. WLOG $u=(1,2)(3,4)$ $v=(1,3,5)$. prove that: for any 3 cycle (i,j,k), we can using conjugate. to change it into $(1,3,5)$.

Step1: $i\to 1$ Step 2: $j\to 3$ Step3: $k\to 5$ (Step by step check)

⚠ Be careful, when changing j,k, you should keep $i=1$ fixed!

$\Rightarrow |Aut(G)| \div |Stab(x)|\,|orb(x)| = |orb(x)| \le |X| = 120$

then $Aut(G) = S_5$ since $S_5 < Aut(G)$. by acting on order 5 element.

A6). $x = (1,2)(3,4)$, $y = (1,2,3)$ $\Rightarrow$ $ord(x)=2$, $ord(y)=3$ $\Rightarrow$ $|\langle x,y\rangle| = |(1,2)(3,4)(1,2,3)| = |(3,2,4)|$

so $ord(xy)=3$.

$X = \{(u,v) \in A_4 \times A_4 \mid ord(u)=2,\ ord(v)=3,\ ord(uv)=3\}$ $\Rightarrow$.

u: 不交的双对换

v: 3-循环.

uv: 3-循环.

$|X| = \frac{1}{2} C_4^2 \times 4 \times 2$

↑ 双对换 ↑ 3-cycle ↑ 2种 $(1,2,3)$ $(1,3,2)$. So $|X| = 24$.

选择时 $(1,3)$ 是同一循环 ↑ 3-cycle $(3,1)$

$Aut(A_4) \curvearrowright X$. ~~$(1,2,4)$~~. then check. $\langle x,y\rangle = A_4$ ↑ generate $(1,2,4)$ $(2,3,4)$ $(1,3,4)$. --

- $Aut(A_4) \curvearrowright X$ free.

A7) $X := \{(u,v) \mid ord(u)=2,\ ord(v)=3,\ ord(uv)=4\}$

- $|Aut(A_4)| \le |X|$ + containing S_4 $\Rightarrow$ $Aut(A_4) = S_4$

v: 3-cycle, uv: 4-cycle $\Rightarrow$ 4 对换 (两奇偶性).

pick $(x,y) \in X$

Claim: $\langle x,y\rangle = S_4$:

Let $x = (1,2)$, then WLOG, $y = (2,3,4)$

$Aut(S_4) \curvearrowright X$ free $\Rightarrow$

Check. $(1,3) = yxy^{-1}$, $(1,4) = y(1,3)y^{-1}$, $(2,3) = x(1,3)x^{-1}$, $(2,4) = x(1,4)x^{-1}$, $(3,4) = (2,3)(2,4)(2,3)$ $\Rightarrow \langle x,y\rangle = S_4$

$|Aut(S_4)| \le |X| = 24$, then. + $S_4 \le Aut(S_4)$. $\Rightarrow$ $Aut(S_4) = S_4$

18). $n \neq 6$. $\phi \in Aut(S_4)$ ϕ send 把对换 映到对换. $\Rightarrow$ ϕ 不改变奇偶性 $\Rightarrow$.

↑ 由讲义之前论

$\phi|_{A_4} \in Aut(A_4)$ somehave injection

$\Rightarrow |Aut(S_4)| \le |Aut(A_4)| = S_4$ $Aut(S_4) \hookrightarrow Aut(A_4)$

since $S_4 \le Aut(S_4)$ ⇓ $Aut(S_4) = S_4$

B A_n $(n \ge 5)$ are simple group

B0) Answer is: A_3: 1, A_3

A_4: 1, $\{1, (1,2)(3,4), (1,3)(2,4), (1,4)(2,3)\}$, A_4.

B1) - $N \lhd A_5$ $N \neq 1$. suppose $N \ni (1,2)(3,4) \in N$ $\Rightarrow$. then. $(2,5)(3,4)\,N\,(2,5)(3,4) = (1,5)(3,4)$

and $\sigma T = (1,2)(3,4)(2,5)(3,4) = (5,1,2)$ is 3 cycle

- $N \lhd A_5$ $N \ni (1,2,3,4,5) \in N$ $\Rightarrow$. let $T = (1,2,3)\,N\,(1,2,3)^{-1}$

$TN^2 = (2,3,1,4,5)(1,2,3,4,5)^2 = (2,3,1)(4,5)(1,2,3,4,5)(1,2,3,4,5)$

$= (1,5,3,2)$

— A_5 has elements of the form

- 3-cycle (3循环)
- 22置换
- 5-循环

} using 1).2) ⇒ if N has element of this type ⇒ has 3-cycle

3-cycle are all conjugate under ~~this~~ conjugate action ⇒ all 3 cycles are in $N \trianglelefteq A_5$ ⇒

$N = A_5$ ⇒ A_5 are simple

B2) · writing down all conjugate classes of A_5

- $\{1\}$
- $(2,2,1)$ · (双对换) · ⟶ · prove that it forms a conjugate class.
- $(3,1,1)$ · $2 \times C_5^3 = 20$

$\frac{1}{2} C_5^2 \cdot C_3^2 = 15$

- (5) · $4! = 24$ ↑ but be careful that: $(i,j,k,l,5)$ ↑ these are not 1 conjugate class in A_5, they are 1 conjugate class in S_5

$g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ i & j & k & l & 5 \end{pmatrix}$

element has the form

then $g(1,2,3,4,5)g^{-1} = (i,j,k,l,5)$

⇓

$\forall \sigma \in X$. if σ conjugate to $(1,2,3,4,5)$ in A_5 ⟺ $g \in A_4$. for corresponding g

⇒ $(1,2,3,4,5)$ has element 12

↑ conjugacy class containing

⇒. A_5 has conjugate class

$\{1\}$	1
$(2,2,1)$	15
$(3,1,1)$	20.
(5) — A	12
(5) — B	12

· — N is union of conjugacy classes

— consider the number divided by 60, only 1, 60 so $|N| = 1$ or 60.

3) · $n \geq 6$ A_{n-1} simple A_n simple. Suppose $N \trianglelefteq A_n$. $|N| \neq 1$ or and $N \neq A_n$

— if $\exists \sigma'$ $\{\sigma'(n) = n\} = N'$ N' be the set of all these σ', then it is a normal

let σ' s.t. $\sigma'(n) = n$ subgrp of A_{n-1} ⇒ · since · $1 \neq \sigma \in N'$ $N' \neq 1$ ⇒ $N' \simeq A_{n-1}$

⇒ N contains all the 3 cycle of the form (i,j,k) where $i,j,k \in n-1$ ⇒ containing all 3-cycle since they are in the same conjugacy class ⇒ $N = A_n$ (since 3-cycles generate A_n).

— $\sigma \in N \setminus \{1\}$ $\{i,j\} \cap \{\sigma(n), n\} = \phi$ ⇒

- $\tau \sigma \tau^{-1} \sigma(n), n$. $\tau = (i,j)(n, \sigma(n))$
- $N \neq A_n \cap \tau \sigma \tau^{-1} \sigma \in N$ ⇒ $\tau \sigma \tau^{-1} \sigma = 1$.

— First. $\forall 1 \leq k \leq n$ $\sigma(k) \neq k$ otherwise let $\sigma(n) = n$ by conjugacy ; so $N = A_n$ ×

$n \geq 6$, can pick $\{k, \sigma(k)\} \cap \{n, \sigma(n)\} = \phi$

pick $i,j \in \{1 \cdots n\} - \{k, \sigma(k), n, \sigma(n)\}$ $\tau = (i,j)(n, \sigma(n))$ ⇒ $\sigma^{-1} \tau \sigma \tau^{-1}(k) = k$

by above ⇒ $\sigma^{-1} \tau \sigma \tau^{-1} = 1$



Then: $\tau\sigma^2\tau^{-1} = (\tau\sigma\tau^{-1})^2 = (\tau\sigma\tau^{-1}\sigma\tau)\cdot(\sigma\tau\cdot\sigma\tau^{-1}) = 1$. $\Rightarrow \sigma^2 = 1 \Rightarrow \sigma$ 为不连对换之积.

if $\sigma(\{i,j\}) \neq \{i,j\} \Rightarrow \sigma$ has decomposition.

$\sigma = (i,s)(j,t)(n,\sigma(n))\cdots \Rightarrow \tau\sigma\tau^{-1}\sigma: i \to j$ then contradiction

which means $\sigma: \{i,j\} \to \{i,j\}$.

B3-4). pick $\{i,j,k\} \cap \{n,\sigma(n)\} = \phi$ then $\sigma: \{i,j\} \to \{i,j\}$.

$\{i,k\} \to \{i,k\}$ $\Rightarrow \sigma(i) = i$

then $\sigma = (n,\sigma(n))$ contradiction

so inductively A_n are simple group for $n \geq 5$

B4) $n \geq 5$ · $G \leq S_n$ + G normal.

$G \cap A_n \lhd A_n$. • Case 1: $A_n \leq G$. $|G| \mid |S_n| \Rightarrow G = A_n$ since $G \neq S_n$.

↑ since. $|S_n|/|A_n| = 2$

$A_n \leq G$ or $G \cap A_n = e$

since A_n is simple. • Case 2: $A_n \cap G = \{e\}$ then all elements are odd

$\Rightarrow$ iff G non-trivial. $\exists \sigma$ $\mathrm{ord}(\sigma) \geq 3 \Rightarrow \sigma^2$ even.

then contradict with $A_n \cap G$.

$\Rightarrow \mathrm{ord}(\sigma) = 2$

$\Rightarrow G$ contains the whole conjugacy class σ belongs to (since $G < S_n$).

suppose σ has the form $(i_1,j_1)\cdots(i_k,j_k)$. then.

has atleast $\dfrac{C_n^2 C_{n-2}^2 \cdots C_{n-2k+2}^2}{k!} \geq 3$ when $n \geq 5$.

pick $\tau_1 \neq \tau_2$ $\sigma\tau_1 = 1$ or even $\Rightarrow$ then since $\tau_1 \neq \tau_2 \Rightarrow$.

in this conjugate class one of $\sigma\tau_1 \cdot \sigma\tau_2$ even $\Rightarrow G \cap A_n = \{e\} \Rightarrow G = \{e\}$

B5) $|S_4| = 24$ $|N| = 4$ $S_4/N = 24/4 = 6$.

~~• $\sigma = (1,2,3)(4,5)$ σ^i is not type $(2,2)$. $\Rightarrow \forall i\ \sigma^i \notin N$ $\mathrm{ord}(\sigma) = 6 \Rightarrow S_4/N = \mathbb{Z}_6$~~

$S_4/N = S_3$ claim: order 6 group has form 1) S_3 check S_4 no order 6

• N is group ✓ 2) $\mathbb{Z}_6$ element $\Rightarrow S_4/N = S_3$

• $\forall \sigma \in S_4$. $\sigma\tau\sigma^{-1}$ has same type $(2,2)$ or identity

both in $N \Rightarrow$. $\sigma\tau\sigma^{-1} \in N$ and $\sigma N\sigma^{-1} = N \Rightarrow N \lhd S_4$.

B6). $H < S_n$ $[S_n : H] = d$. $N \lhd A_4$ since all element are in A_4

$S_n \curvearrowright S_n/H$ cosets $S_n/H = \{a_1H \cdots a_dH\}$

↑ set of cosets of H

$S_n \longrightarrow S_d$. and $\ker\varphi$.

$\varphi: \sigma \longmapsto (a_iH \to \sigma a_iH) \Rightarrow \sigma \in \ker\varphi \Rightarrow a_iH = \sigma a_iH \Rightarrow a_i^{-1}\sigma a_i \in H$

$\Rightarrow \sigma \in a_iHa_i^{-1} \overset{\forall i \leq i \leq d}{\Leftrightarrow} \sigma \in (a_iH)H(a_iH)^{-1}$

$\Leftrightarrow \sigma \in \bigcap_{g \in G} gHg^{-1}$

and $\bigcap_{g \in G} gHg^{-1} < H \Rightarrow \ker\varphi < H$

$\forall g' \in S_n$ $g'(\bigcap_{g \in G} gHg^{-1})(g')^{-1} = \bigcap_{g \in G} gHg^{-1} \Rightarrow \bigcap_{g \in G} gHg^{-1} \lhd S_n$

$H \neq A_n, S_n \Rightarrow \bigcap_{g \in G} gHg^{-1} = \{e\}$

(since if $H \neq A_n, S_n$ $\bigcap_{g \in G} g^{-1}Hg \neq H$

$\Rightarrow \bigcap_{g\in G} gHg^{-1} = \{e\} \Rightarrow \ker\varphi = \{e\} \Rightarrow |S_d| \geq |S_n|$

$\Rightarrow d \geq n$

B7). $n \geq 2$ $S_n \longrightarrow A_{n+2}$.

$\cdot \varphi(\sigma) = \begin{cases} \sigma & \sigma \text{ even} \\ \sigma(n+1, n+2) \cdot \sigma & \sigma \text{ odd.} \end{cases}$ · check: it is group hom.

· If $\exists \varphi: S_n \hookrightarrow A_{n+2} \Rightarrow$ $A_{n+2} \curvearrowright A_{n+2}/\mathrm{Im}\varphi$ (← set of cosets) $\Rightarrow |A_{n+2}/\mathrm{Im}\varphi| = \frac{n+2}{2}$

if $|S_n| \hookrightarrow A_{n+2} \Rightarrow \mathrm{Im}\varphi \simeq S_n$. $\{a_1\mathrm{Im}\varphi \cdots a_k\mathrm{Im}\varphi\}$

$\Rightarrow \frac{|A_{n+2}|}{|S_n|} = \frac{n+2}{2} \Rightarrow 2 \mid n+2$. then $A_{n+2} \xrightarrow{\varphi} S_{|A_{n+2}/\mathrm{Im}\varphi|}$

but $|A_{n+2}| = \frac{(n+2)!}{2}$ $S_{|A_{n+2}/\mathrm{Im}\varphi|} = \left(\frac{n+2}{2}\right)!$

$\Rightarrow |\ker\varphi \neq \{e\}$ but A_{n+2} simple

$\Rightarrow \ker\varphi = A_{n+2} \Rightarrow \forall \sigma \in A_{n+2}\ \sigma\mathrm{Im}\varphi = \mathrm{Im}\varphi$

$\Rightarrow \sigma \in \mathrm{Im}\varphi$

$\Rightarrow$ $A_{n+2} \subseteq \mathrm{Im}\varphi$ but $|\mathrm{Im}\varphi| = n!$

$|A_{n+2}| = \frac{(n+2)!}{2}$ $\Rightarrow$ Contradiction

so $\nexists\ S_n \hookrightarrow A_{n+2}$ monomorphism ×

↑ when $n \geq 5$ (using A_n simple)

↓ $n=4$ $S_3 \hookrightarrow A_4$

↑ order 2 element ↑ order 2 element $(1,2)(3,4)$

$(1,3)(2,4)$

$(1,4)(2,3)$

$(1,2)(1,3)(2,3)$

so in conclusion: no $S_n \hookrightarrow A_{n+2}$ injective $\forall n$

if σ exist $(1,2)(1,3) = (3,2,1)$ ← order 3

$(1,2)(3,4)(1,3)(2,4) = (4,1)(2,3)$

↑ order 2 it fails

C. $PSL(2;\mathbb{F}_2) \overset{\simeq S_3}{\simeq} PSL(2;\mathbb{F}_3) \simeq A_4$.

C1). $GL(2;\mathbb{F}_2)$. $\sigma \in GL(2;\mathbb{F}_2) \Rightarrow \det(\sigma) \neq 0 \overset{\text{in } \mathbb{F}_2}{\Longrightarrow} \det(\sigma) = 1$

$\Rightarrow \sigma \in SL(2;\mathbb{F}_2) \Rightarrow \sigma \in PSL(2;\mathbb{F}_2)$

$k^\times \otimes I = I$ in $\mathbb{F}_2$ $(k^\times = \{1\})$

then $GL(2;\mathbb{F}_2) \simeq SL(2;\mathbb{F}_2) \simeq PGL(2;\mathbb{F}_2) \simeq PSL(2;\mathbb{F}_2)$

$PSL(n;k) = SL(n;k)/\mu_{n+1}(k)\cdot I_{n+1}$

By definition:

C2) $GL(2;\mathbb{F}_2)$ ↑ 2×2-matrix. $\mathbb{P}^1(\mathbb{F}_2)$ $\begin{pmatrix}1\\0\end{pmatrix} \to$ ∉ $\mathbb{P}^1(\mathbb{F}_2) = \left\{\begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}\right\}$

$GL(2;\mathbb{F}_2) \curvearrowright \mathbb{P}^1(\mathbb{F}_2) \Rightarrow GL(2;\mathbb{F}_2) \longrightarrow S_{\mathbb{P}^1(\mathbb{F}_2)} = S_3$

· $|GL(2;\mathbb{F}_2)| = (2^2-2)(2^2-1) = 6$ $|S_3| = 3! = 6$

C3). $\mathbb{P}^1(\mathbb{F}_3) = \left\{\begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1\\2\end{pmatrix}\begin{pmatrix}0\\1\end{pmatrix}\right\}$ (OU) direct check

· $PSL(n+1;k) \curvearrowright \mathbb{P}^n(k)$. $\Rightarrow GL(2;\mathbb{F}_2) \simeq PSL(2;\mathbb{F}_2) \hookrightarrow S_3$

$\Rightarrow PSL(2;\mathbb{F}_2) = GL(2;\mathbb{F}_2) = S_3$

$\Rightarrow$ $GL(2;\mathbb{F}_3) \curvearrowright \mathbb{P}^1(\mathbb{F}_3)$. $\Rightarrow GL(2;\mathbb{F}_3) \overset{\varphi}{\hookrightarrow} S_4 = S_{|\mathbb{P}^1(\mathbb{F}_3)|}$

$\Rightarrow \ker\varphi = \mu_2(\mathbb{F}_3)\,id_V$ $\Rightarrow$ $PGL(2;\mathbb{F}_3) = \frac{(3^2-3)(3^2-1)}{2} = 24$

$\mu_2(\mathbb{F}_3) = \{1, 2\}$.

$\Rightarrow$ $|PGL(2;\mathbb{F}_3) \simeq S_4 = S_{|\mathbb{P}^1(\mathbb{F}_3)|}$

C4).

Claim $[S_n : G] = 2 \Rightarrow G \simeq A_n$.

$\varphi: GL(2:\mathbb{F}_3) \longrightarrow S_{\mathbb{P}^1(\mathbb{F}_3)} \simeq S_4$ $\quad \psi: SL(2:\mathbb{F}_3) \longrightarrow S_{\mathbb{P}^1(\mathbb{F}_3)} \simeq S_4$ $\quad \psi = \varphi|_{SL(2:\mathbb{F}_3)}$

$\ker\varphi \subseteq SL(2:\mathbb{F}_3)$.

$\{1 \cdot id_V, -1 \cdot id_V\}$ $\quad |SL(2:\mathbb{F}_3)| = \frac{1}{2}|GL(2:\mathbb{F}_3)| = 12$

$|PSL(2:\mathbb{F}_3)| = \frac{\#SL(2:\mathbb{F}_3)}{\#\ker\varphi}$ since $PSL(2:\mathbb{F}_3) = GL(2:\mathbb{F}_3)/\mathbb{F}_3^\times$.

$\Rightarrow [S_4 : \mathrm{Im}\psi] = 2 \Rightarrow \mathrm{Im}\psi = A_4 \Rightarrow PSL(2:\mathbb{F}_3) \simeq A_4$.

pf of claim: Step 1: $[S_n : G] = 2 \Rightarrow G$ normal.

2: Apply B4.

C5) $\mathbb{P}^1(\mathbb{F}_4) = \{$  $\}$ 5 elements.

$\mathbb{F}_4 \neq \mathbb{Z}/4\mathbb{Z}$! $\quad \forall x \in \mathbb{F}_4$, $1 = -1$, $2x = 0$.

$\Rightarrow PSL(2:\mathbb{F}_4) \hookrightarrow S_5$ $\quad$ Aim: $SL(2:\mathbb{F}_4) \cap \mu_2(\mathbb{F}_4)\, id_V = \{id_V\}$.

i.e. $a \in \mathbb{F}_4$, $a^2 = 1 \Rightarrow a = \pm 1$

so. $SL(2:\mathbb{F}_4) \simeq PSL(2:\mathbb{F}_4)$. (direct check)

$\Rightarrow \varphi: PSL(2:\mathbb{F}_4) \hookrightarrow S_5$ (is $SL(2:\mathbb{F}_4)$) $\Downarrow$

$|\mathrm{Im}\varphi| = |PSL(2:\mathbb{F}_4)| = |SL(2:\mathbb{F}_4)| = \frac{(4^2-1)(4^2-4)}{3} = \frac{15\times 12}{3} = 60$

$\Rightarrow \mathrm{Im}\varphi \simeq A_5 \Rightarrow PSL(2:\mathbb{F}_4) \simeq A_5$

C6). First: $H < S_n$ ordern:

$\Rightarrow S_n \curvearrowright S_n/H \Rightarrow S_n \xrightarrow{\varphi} S_{|S_n/H|}$ ↓ from B6 φ injective

$\Rightarrow |S_n| = |S_{|S_n/H|}| = n! \Rightarrow \varphi$ isom.

$H \curvearrowright S_n/H \Rightarrow H \curvearrowright \{a_1H, \dots a_nH\}$ fixing $H \Rightarrow |\mathrm{Im}\psi| = |H| = (n-1)!$

and injective $\Rightarrow H \hookrightarrow S_{n-1} \Rightarrow H = S_{n-1}$ ($|H| = |S_{n-1}|$)

$|PSL_2(\mathbb{F}_5)|$

$GL(2:\mathbb{F}_5) \xrightarrow{\varphi} S_6 = S_{\mathbb{P}^1(\mathbb{F}_5)}$. $\Rightarrow |PGL(2:\mathbb{F}_5)| = |\mathrm{Im}\varphi| = \frac{|GL(2,\mathbb{F}_5)|}{|\ker\varphi|}$ Since using $a_1H \dots a_nH$, $a_iH = H$ $a_i \longmapsto i$ then H view as S_{n-1}.

$\ker\varphi = \{\pm k\, id_V \mid k = 1 \dots 4\}$

$= \frac{(5^2-1)(5^2-5)}{4}$

$= 6 \times 20 = 120$

$\Rightarrow [S_6 : \mathrm{Im}\psi] = 6 \Rightarrow \mathrm{Im}\psi \simeq S_5 \Rightarrow PGL(2:\mathbb{F}_5) \simeq \mathrm{Im}\psi \simeq S_5$

C7) (non-example)

$-\alpha^2 = id_V$ $\alpha \in SL(2;\mathbb{F}_q) \Rightarrow (\alpha - id_V)(\alpha + id_V) = 0 \Rightarrow (\alpha \pm id_V)^2 = 0$ since $2 = 0$ in $\mathbb{F}_q$

$\Rightarrow \dim\ker(\alpha \pm id_V) \geq 2$. $\Rightarrow$ 剪切映射

$\Rightarrow$. 仅一个共轭类

↑ 剪切映射均相互共轭

Step2:

• $\varphi: SL(3:\mathbb{F}_4) \longrightarrow PSL(3:\mathbb{F}_4)$. $\sigma^2 = e$.

$\sigma \longmapsto \varphi(\sigma)$. $\varphi(\sigma)^2 = a$ but $\varphi(\sigma) \neq e$. $\operatorname{ord}(\varphi(\sigma)) = 2$.

↑ otherwise. $\sigma = c \cdot id_V \Rightarrow \operatorname{ord}(\sigma) \neq 2$ ($c \in \mathbb{F}_4^\times$ → check.)

Aim • $\varphi|$ ord 2 elements in $SL(3:\mathbb{F}_4)$. injective.

surjective

if not: $\forall \sigma, \tau \in SL(3:\mathbb{F}_4)$

$\varphi(\sigma) = \varphi(\tau) \Rightarrow \sigma\tau^{-1} = c\,id \Rightarrow$ by eigenvalue $= 1$

$\Rightarrow \sigma\tau^{-1} = id_V \Rightarrow \sigma = \tau \Rightarrow$ inj

surj: $\forall \tau \in PSL(3:\mathbb{F}_4)$ $\forall \sigma \in \varphi^{-1}(\tau)$

$\sigma^2 = c\,id_V$ ($c \in \mathbb{F}_4^\times$). $\sigma \neq c'\,id_V$) $\Rightarrow$ $\exists s$ $s^2 = c$. s and $\frac{\sigma}{s}$ not has the form $c' \cdot id_V$.

$\sigma^2 = id_V$

$\operatorname{ord}\left(\frac{\sigma}{s}\right) = 2$. → direct check

$\mathbb{F}_4 = \{0, 1, a, 1-a\}$ $\sigma^2 = a\,id_V$ or $(1-a)\,id_V$: ~~contradict~~.

means. $\forall$ ord 2 element in $PSL(3:\mathbb{F}_4)$

φ: {ord 2 ele in $SL(3:\mathbb{F}_4)$} so $\operatorname{ord}(\sigma) \neq 2$. can pick $\sigma \in \varphi^{-1}(\tau)$ s.t. $\operatorname{ord}(\varphi^{-1}(\tau))$

Step3: ⟷ {ord 2 ele in $PSL(3:\mathbb{F}_4)$} ~~$\varphi^{-1}(\tau)$ correspond to it~~

$A_8 \not\cong PSL(3:\mathbb{F}_4)$

↑ ⟵ by step 1 + step 2, only 1 conjugacy class.

(2,2,2,2)
(2,2,1,1,1,1) } ⇒ 2 type of ord 2 element conjugacy class

• $|PSL(3:\mathbb{F}_4)| = \frac{(4^3-1)(4^3-4)(4^3-4^2)}{3 \times 3}$

$= 7 \times 60 \times 48$

$|A_8| = \frac{8!}{2} = 20160$

$= 28 \times 720 = 20160$

• Simple : A_8 from B3)

$PSL(3:\mathbb{F}_4)$ from notes (Iwasawa criterion).